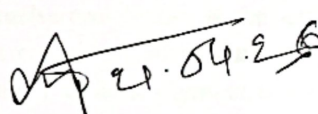




DAFFODIL INTERNATIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FYDP (TITLE PHASE) EVALUATION REPORT

REPORTING PERIOD- SPRING-2026

Project Identification:

I. Project Title	The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening.	
II. Group Members	ID: 0242310005101484	Name: MD. Shoaib Khan
	ID: 02423100051011412	Name: Asmita Rahman
III. Supervisor	Name: Dr. Md. Ali Hossain	Designation: Associate Professor
IV. Co-Supervisor	Name: Mayen Uddin Mojumder	Designation: Assistant Professor
V. Submission Date:	21-04-2026	
VII. Certificate:	"This is to certify that the final year design project work until Title defense evaluation held on _____, titled as stated in <i>Sec. I</i>, executed by the students group mentioned in <i>Sec. II</i>, have been found satisfactory and every section of this report is reflecting the same."	 (Signature of Supervisor & date)

CO Description and Corresponding PO for Title Defense Phase

CO	CO Descriptions	PO
CO1	"Synthesize recently acquired and pre-existing engineering knowledge to critically identify and define a real-world complex engineering challenge in AI-based multi-modal medical diagnostics, while explaining the fundamental concepts of foundation models, multi-modal medical imaging, and pan-organ diagnostic systems in the context of AI-driven healthcare screening, as addressed through the development of 'The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening.' "	PO1
CO2	Examine the core engineering objectives and architectural design considerations of an AI-driven multi-organ diagnostic foundation model to formulate effective solutions for 'The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening'	PO2
CO3	Investigate relevant literature in medical AI and multi-modal imaging to identify research gaps, outline problem boundaries, and define measurable targets for the FYDP of 'The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening'	PO4

1. Introduction:

The rapid advancement of artificial intelligence has opened transformative possibilities in the field of medical diagnostics, enabling systems to process and interpret complex clinical data with remarkable accuracy. Traditional diagnostic approaches are often limited to single-organ analysis, requiring multiple specialized tools and significant clinical expertise, which increases both time and cost in healthcare delivery. There is a growing demand for intelligent, unified diagnostic frameworks capable of screening multiple organs simultaneously across diverse imaging modalities.

This project proposes **"The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening"** — an AI-driven foundation model designed to

address this critical gap. The system aims to integrate multi-modal medical imaging data, including MRI, CT, and X-ray, into a single high-capacity model capable of performing simultaneous pan-organ disease screening. By leveraging state-of-the-art deep learning architectures and large-scale medical datasets, the proposed model seeks to enhance diagnostic accuracy, reduce clinical workload, and improve early disease detection across multiple organ systems.

This Final Year Design Project represents a significant step toward building intelligent, scalable, and clinically viable AI solutions for next-generation healthcare screening systems.

2. Background Study:

The rapid proliferation of AI-powered diagnostic tools has marked a paradigm shift in modern healthcare, enabling automated analysis of complex medical imaging data with unprecedented speed and accuracy. Conventional deep learning models, built upon architectures such as ResNet, EfficientNet, DenseNet, and Vision Transformers, have individually demonstrated strong diagnostic performance across isolated imaging tasks. Transfer learning has further enhanced their clinical applicability, allowing models pretrained on large-scale datasets to be fine-tuned for specific medical conditions even under limited data availability.

However, the current state of AI-driven medical diagnostics reveals several deeply rooted limitations that hinder large-scale clinical deployment:

- **Organ-specific constraints** — Existing models are predominantly designed for single-organ analysis, unable to generalize across diverse anatomical structures
- **Modality limitations** — Most frameworks operate on a single imaging modality such as MRI, CT, or X-ray, failing to leverage the complementary diagnostic value of multi-modal data fusion
- **Absence of unified pipelines** — No clinically deployed system currently integrates pan-organ screening within a single high-capacity diagnostic framework
- **Weak interpretability** — Limited adoption of explainable AI mechanisms reduces physician trust and hampers clinical acceptance
- **Insufficient validation** — Lack of rigorous statistical evaluation across heterogeneous patient populations restricts real-world reliability

These shortcomings are particularly significant given the realities of clinical practice. In actual diagnostic workflows, radiologists and clinicians must simultaneously evaluate multiple organ systems across diverse imaging modalities — a cognitively demanding task that demands intelligent, unified decision-support systems capable of matching this breadth.

Foundation models, originally developed for natural language processing and later extended to computer vision, have recently emerged as a transformative approach in medical AI. Their ability to learn generalizable representations from massive, heterogeneous datasets positions them as ideal candidates for building comprehensive, multi-organ diagnostic systems. Recent works such as MedSAM, UniverSeg, and BioViL have demonstrated the viability of adapting large pretrained models to medical imaging tasks, yet a truly high-capacity, pan-organ, multi-modal screening model remains an unaddressed research frontier.

Multi-modal medical imaging encompassing MRI, CT, PET, and X-ray captures complementary physiological and anatomical information across organ systems including the brain, lungs, liver, kidneys, and cardiovascular structures. Effective fusion of these modalities within a unified architectural framework presents both a significant technical challenge and a critical clinical opportunity. Addressing this challenge is central to the motivation of this project.

This project, **"The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening,"** directly responds to these gaps by proposing a scalable, unified AI framework capable of simultaneous multi-organ disease screening across heterogeneous imaging modalities. By integrating foundation model architectures with multi-modal data fusion strategies and explainable AI techniques, this work aims to bridge the divide between narrow AI diagnostic tools and a clinically viable, comprehensive screening paradigm representing a significant contribution to the advancement of AI-driven healthcare.

3. Problem Statement:

Despite significant progress in AI-driven medical diagnostics, existing systems remain fundamentally limited in their ability to address the full complexity of real-world clinical screening requirements. Current deep learning-based diagnostic models are predominantly designed for single-organ, single-disease, or single-modality analysis, creating a fragmented diagnostic landscape that fails to replicate the comprehensive evaluative capacity of trained clinical professionals.

In practical healthcare environments, radiologists and diagnostic specialists must simultaneously assess multiple organ systems across diverse imaging modalities — including MRI, CT, X-ray, and PET scans — to arrive at accurate and holistic clinical conclusions. The absence of a unified, high-capacity AI framework capable of performing such pan-organ, multi-modal screening introduces critical inefficiencies in diagnostic workflows, increases the risk of missed or delayed diagnoses, and places an unsustainable cognitive burden on healthcare professionals operating under resource-constrained conditions.

The core problems addressed by this project are:

- **Fragmented diagnostic AI** — Existing models target isolated conditions or organs, lacking the architectural capacity to generalize across diverse pathological and anatomical domains simultaneously
- **Multi-modal data gap** — Current frameworks fail to effectively fuse complementary diagnostic information from heterogeneous imaging modalities, limiting overall diagnostic comprehensiveness
- **Absence of a pan-organ framework** — No clinically viable, unified foundation model exists that can screen multiple organ systems within a single end-to-end inference pipeline
- **Limited explainability** — The lack of transparent AI decision mechanisms reduces clinical trust, hindering the adoption of AI tools in critical diagnostic settings
- **Scalability constraints** — Existing diagnostic models lack the capacity to scale across diverse patient populations, disease categories, and imaging protocols required for large-scale clinical deployment

These limitations collectively underscore the urgent need for a paradigm shift in medical AI — one that moves beyond narrow, task-specific models toward a generalized, high-capacity diagnostic foundation model. This project addresses this critical gap through the development of **"The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening"** — an integrated AI framework designed to deliver accurate, explainable, and scalable diagnostic screening across multiple organ systems and imaging modalities, aligned with the demands of modern clinical practice.

4. Preliminary Novelty Statement:

The proposed project "**The Pan-Organ Diagnostic Paradigm: A High-Capacity Foundation Model for Multi-Modality Medical Screening**" introduces the following novel contributions to AI-driven medical diagnostics:

- **Pan-Organ Diagnostic Architecture** — A unified foundation model capable of simultaneously screening multiple organ systems within a single end-to-end inference pipeline, moving beyond narrow organ-specific frameworks.
- **Multi-Modal Data Fusion** — Seamless integration of heterogeneous imaging modalities including MRI, CT, and X-ray, leveraging complementary diagnostic information to enhance screening accuracy.
- **High-Capacity Foundation Model** — Application of large-scale pretrained representations specifically tailored for pan-organ medical screening, overcoming the scalability constraints of conventional diagnostic models.
- **Comprehensive Disease Coverage** — A clinically structured taxonomy spanning neurodegenerative, neoplastic, vascular, demyelinating, infectious, traumatic, epileptic, and autoimmune conditions within a single unified model.
- **Explainable AI Integration** — Incorporation of gradient-weighted class activation mapping to generate visual diagnostic explanations, improving clinical transparency and physician trust.

Collectively, these contributions establish this project as a significant step toward a generalized, interpretable, and clinically scalable AI diagnostic paradigm for next-generation medical screening.

5. Research Questions:

How can the proposed framework be evaluated and validated against established clinical benchmarks to demonstrate its reliability, scalability, and suitability for real-world healthcare deployment?

6. Objectives:

The primary objective of this project is to design and develop a high-capacity AI foundation model capable of performing accurate, explainable, and scalable pan-organ disease screening across heterogeneous medical imaging modalities. The specific objectives are as follows:

- **O1: Literature Review and Problem Analysis** To conduct a comprehensive review of existing AI diagnostic frameworks, foundation model architectures, and multi-modal medical imaging techniques in order to identify critical research gaps and establish a strong theoretical foundation for the proposed system
- **O2: Dataset Acquisition and Preprocessing** To collect, curate, and preprocess multi-modal medical imaging datasets encompassing MRI, CT, and X-ray modalities across multiple organ systems, ensuring data quality, diversity, and clinical relevance for model training and evaluation
- **O3: Foundation Model Architecture Design** To design and develop a high-capacity deep learning architecture capable of simultaneously processing heterogeneous imaging modalities and generalizing across diverse organ structures, disease categories, and imaging protocols within a unified diagnostic pipeline
- **O4: Multi-Modal Data Fusion Implementation** To implement and optimize effective multi-modal data fusion strategies that leverage complementary diagnostic information from MRI, CT, and X-ray inputs, maximizing the overall diagnostic comprehensiveness and accuracy of the proposed framework
- **O5: Explainable AI Integration** To integrate gradient-weighted class activation mapping and related interpretability mechanisms into the proposed model, generating spatially localized visual explanations that enhance clinical transparency and support physician decision-making
- **O6: Model Training and Optimization** To train, fine-tune, and optimize the proposed foundation model using transfer learning strategies and large-scale pretraining techniques, ensuring high diagnostic performance across diverse patient populations and disease categories under constrained dataset conditions
- **O7: Performance Evaluation and Validation** To rigorously evaluate and validate the proposed model against established clinical benchmarks and standard performance metrics including accuracy, sensitivity, specificity, and AUC-ROC, demonstrating its reliability and suitability for real-world healthcare deployment

7. Expected Outcome:

- 1. A unified high-capacity foundation model capable of simultaneous pan-organ disease screening across multiple medical imaging modalities**

2. A robust multi-modal fusion framework effectively integrating MRI, CT, and X-ray data for comprehensive diagnostic output
3. Improved diagnostic accuracy, sensitivity, and specificity across diverse organ systems compared to existing single-organ AI models
4. Clinically interpretable heatmap visualizations highlighting pathologically relevant regions to enhance physician trust and transparency
5. Comprehensive disease taxonomy coverage spanning neurodegenerative, neoplastic, vascular, demyelinating, infectious, traumatic, and autoimmune conditions
6. Identification of the optimal foundation model architecture and fusion strategy for large-scale pan-organ screening
7. A scalable and generalizable diagnostic system validated across diverse patient populations and heterogeneous imaging protocols
8. A clinically deployable AI screening tool capable of reducing diagnostic workload and supporting real-world clinical decision-making
9. A meaningful academic contribution establishing a foundational benchmark for future multi-modal, multi-organ diagnostic AI research

8. Scope of Work:

Included:

1. Pan-organ disease screening across multiple organ systems using MRI, CT, and X-ray imaging modalities
2. High-capacity foundation model development for simultaneous multi-organ, multi-disease diagnostic classification
3. Multi-modal data fusion strategies for enhanced diagnostic comprehensiveness and accuracy
4. Transfer learning and large-scale pretraining techniques for optimized model performance
5. Explainable AI integration using gradient-weighted class activation mapping for clinical interpretability
6. Rigorous performance evaluation against established clinical benchmarks and standard metrics

Limitations:

1. Dependent on availability and diversity of publicly accessible multi-modal medical imaging datasets
2. Coverage of extremely rare diseases may be limited due to insufficient training data
3. Clinical validation restricted to experimental benchmark datasets, excluding real-time hospital deployment

9. References:

- [1] J. Ma et al., "Segment Anything in Medical Images," *Nature Communications*, vol. 15, no. 654, Jan. 2024. <https://doi.org/10.1038/s41467-024-44824-z>
- [2] K. Huemann et al., "Multimodal Foundation Models for Medical Imaging — A Systematic Review and Implementation Guidelines," *medRxiv*, Oct. 2024. <https://doi.org/10.1101/2024.10.23.24316003>
- [3] M. R. H. Hosseinzadeh Taher et al., "Learning Foundation Models from Multi-Organ Medical Images by Capturing Consistency and Diversity of Anatomical Structures," *Springer Nature*, 2024.
- [4] Y. Zhang et al., "Medical Multimodal Foundation Models in Clinical Diagnosis and Treatment: Applications, Challenges, and Future Directions," *arXiv*, Dec. 2024. <https://arxiv.org/abs/2412.02621>
- [5] H. Chen et al., "Multi-Organ Foundation Model for Universal Ultrasound Image Segmentation With Task Prompt and Anatomical Prior," *IEEE Transactions on Medical Imaging*, vol. 44, no. 2, pp. 1005–1018, Feb. 2025.
- [6] B. D. Lee et al., "A Narrative Review of Foundation Models for Medical Image Segmentation: Zero-Shot Performance Evaluation on Diverse Modalities," *Quantitative Imaging in Medicine and Surgery*, vol. 15, no. 6, pp. 5825–5858, Jun. 2025.
- [7] S. Ye et al., "Foundation Models Meet Medical Image Interpretation," *Research*, Feb. 2026. <https://doi.org/10.34133/research.1024>
- [8] W. Moor et al., "Foundation Models for Generalist Medical Artificial Intelligence," *Nature*, vol. 616, pp. 259–265, 2023.

- [9] R. Bommasani et al., "On the Opportunities and Risks of Foundation Models," *arXiv*, Aug. 2021. <https://arxiv.org/abs/2108.07258>
- [10] S. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," *International Journal of Computer Vision*, vol. 128, pp. 336–359, 2020.

Supervisor's Feedback:

Provide comments and suggestions for improvement

1. more literature review
 2. dataset collection
 3. data preprocessing
- gap finding
method development

Examiner's Signature: _____

Date: 21.04.2026